

Technical Specifications

Types	Flat (DIN 125), Spring (DIN 127), Lock (DIN 6798), Belleville (DIN 2093), Fender, Structural (ASTM F436)
Size Range	M3 - M100 metric; 3/16"-4" imperial; custom OD/ID available
Standards	DIN 125-A/B, DIN 127-A/B, DIN 137, DIN 2093, ISO 7089, ISO 7090, ASTM F436, BS 4320, IS 2016
Materials	Carbon Steel (MS), SS 304 (A2), SS 316 (A4), Alloy Steel, Brass, Copper, Bronze
Finishes	Plain, Zinc Electroplated, Hot-Dip Galvanized (HDG), Black Oxide, Phosphate, Dacromet
Thickness	Per DIN/ISO standard; custom 0.5 mm - 20 mm available
MOQ	10,000 pieces or 700 kg
Lead Time	5-14 working days (standard); 14-21 days custom dimensions
Certification	Mill Test Certificate (EN 10204 Type 3.1) - Optional SGS / BV / TUV inspection
HS Code	7318.22

Grades / Types at a Glance

Washer Type	Standard	Primary Function
Flat Washer	DIN 125 / ISO 7089	Distributes load, protects surface
Spring Washer	DIN 127	Resists loosening under vibration
Lock Washer	DIN 6798	Teeth bite surface, prevents rotation
Belleville Disc Spring	DIN 2093	High-load preload maintenance
Fender Washer	DIN 9021	Large OD for soft or oversized holes
Structural / Hardened	ASTM F436	For A325 / A490 structural bolts

Key Applications

- Structural steel bolted connections & fabrication
- Pipe flanges, pressure vessels, and heat exchangers
- Automotive, rail, and heavy vehicle assemblies
- Vibration-prone machinery and rotating equipment
- Electrical switchgear, panel boards, cable management
- HVAC, plumbing, and building services installations

Export & Quality Information

Export Terms	FOB / CIF - Chennai, Tuticorin, Mumbai, Mundra
Currencies	USD / EUR / AED
MOQ	10,000 pieces or 700 kg
HS Code	7318.22
Quality / Cert	Mill Test Certificate (EN 10204 Type 3.1) - Optional SGS / BV / TUV inspection